

Cauchy - Green Strain Tensor

For $\underline{x} = \varphi(\underline{X})$ with $\underline{F} = \nabla \varphi$

$$\underline{C} = \underline{F}^T \underline{F} = \underline{U}^2$$

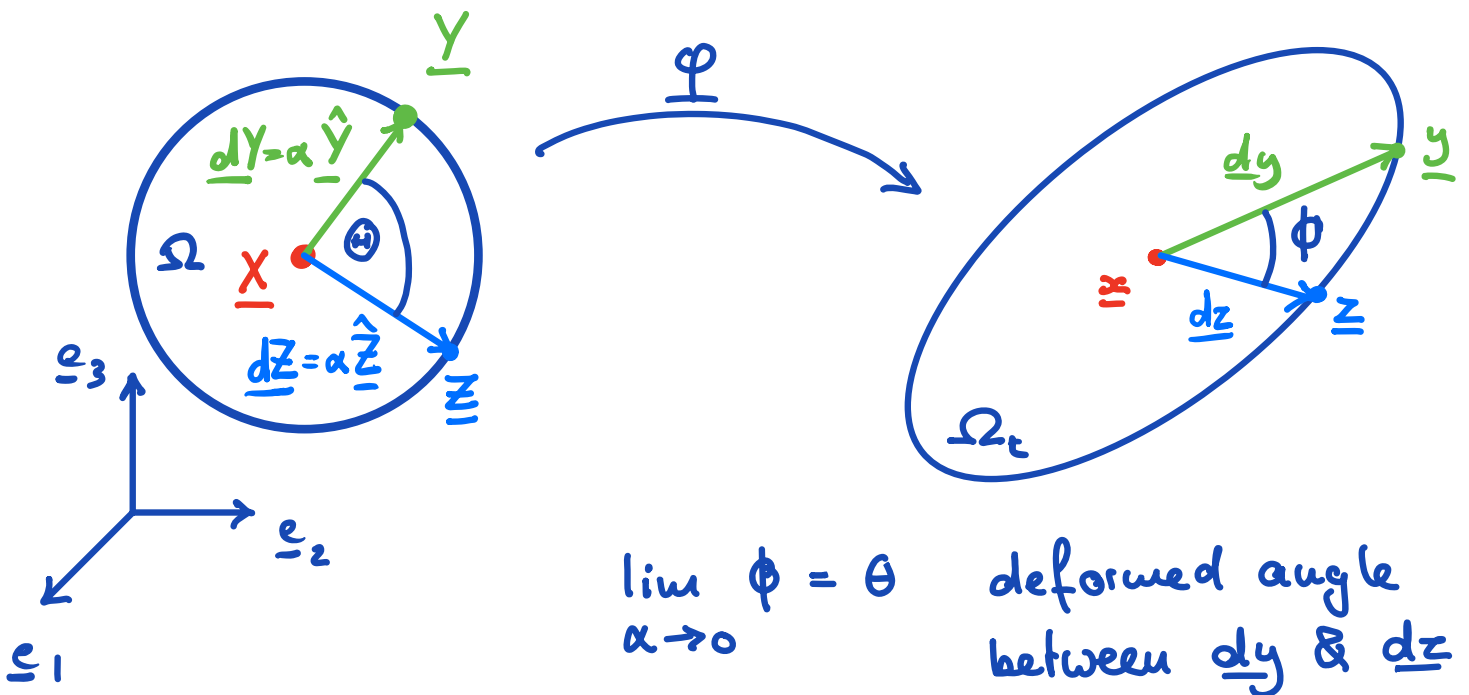
sym. pos. def.

$\underline{U}$ is right-stretch tensor $\underline{F} = \underline{R} \underline{U}$

$\Rightarrow$ only information about stretches

Interpretation of $\underline{C}$

How are changes in relative position and orientation of material points quantified by $\underline{C}$?



Cauchy - Green strain relations

For any point $\underline{X} \in B$ and unit vectors $\hat{\underline{Y}}$ and $\hat{\underline{Z}}$ we define $\lambda(\hat{\underline{Y}}) > 0$ and $\theta(\hat{\underline{Y}}, \hat{\underline{Z}}) \in [0, \pi]$ by

$$\lambda(\hat{\underline{Y}}) = \sqrt{\hat{\underline{Y}} \cdot \underline{\underline{C}} \hat{\underline{Y}}}$$

and

$$\cos \theta(\hat{\underline{Y}}, \hat{\underline{Z}}) = \frac{\hat{\underline{Y}} \cdot \underline{\underline{C}} \hat{\underline{Z}}}{\sqrt{\hat{\underline{Y}} \cdot \underline{\underline{C}} \hat{\underline{Y}}} \sqrt{\hat{\underline{Z}} \cdot \underline{\underline{C}} \hat{\underline{Z}}}}$$

I. Stretches

1D: $\lambda = \frac{\ell}{L}$ ratio of deformed to initial length

3D: $\lambda(\hat{\underline{Y}}) = \frac{|\underline{dy}|}{|\underline{dY}|}$ stretch in direction $\hat{\underline{Y}}$ at $\underline{X}$.

To determine the stretch we use $\underline{dy} = \underline{\underline{F}}(\underline{X}) \underline{dY}$.

$$\begin{aligned} |\underline{dy}|^2 &= \underline{dy} \cdot \underline{dy} = \underline{\underline{F}} \underline{dY} \cdot (\underline{\underline{F}} \underline{dY}) = \underline{dY} \cdot \underline{\underline{F}}^T \underline{\underline{F}} \underline{dY} = \underline{dY} \cdot \underline{\underline{C}} \underline{dY} \\ &= \alpha^2 \hat{\underline{Y}} \cdot \underline{\underline{C}} \hat{\underline{Y}} \end{aligned}$$

$|\underline{dY}|^2 = \alpha^2$ by definition

$$\Rightarrow \lambda^2(\hat{\underline{Y}}) = |\underline{dy}|^2 / |\underline{dY}|^2 = \hat{\underline{Y}} \cdot \underline{\underline{C}} \hat{\underline{Y}}$$

so that $\lambda(\hat{\underline{Y}}) = \sqrt{\hat{\underline{Y}} \cdot \underline{\underline{C}} \hat{\underline{Y}}}$ ✓

If $\underline{u}_p$ is a right-principal stretch, so that

$$(\underline{C} - \lambda_p^2 \underline{I}) \underline{u}_p = \underline{0} \quad (\text{no sum})$$

$$\underline{u}_p \cdot (\underline{C} - \lambda_p^2 \underline{I}) \underline{u}_p = 0$$

$$\underline{u}_p \cdot \underline{C} \underline{u}_p - \lambda_p^2 \underline{u}_p \cdot \underline{u}_p = 0$$

$$\underline{u}_p \cdot \underline{C} \underline{u}_p = \lambda_p^2$$

$$\Rightarrow \boxed{\lambda_p = \sqrt{\underline{u}_p \cdot \underline{C} \underline{u}_p}} \equiv \lambda(\underline{u}_p) \quad |\underline{u}_p| = 1$$

justifies referring to λ_i 's as principal stretches.

Arguments similar to determination of principal stresses show that $\lambda(\hat{Y})$ has extremum if $\hat{Y} = \hat{u}_i$.

II. Shear

Change in angle

$$\boxed{\gamma(\hat{Y}, \hat{Z}) = \Theta(\hat{Y}, \hat{Z}) - \theta(\hat{Y}, \hat{Z})}$$

$\Theta(\underline{\hat{Y}}, \underline{\hat{Z}})$ angle between $\underline{\hat{Y}}$ & $\underline{\hat{Z}}$ in initial conf.

$\theta(\underline{d\hat{Y}}, \underline{d\hat{Z}})$ angle between $\underline{dy}$ & $\underline{dz}$ in limit $\alpha \rightarrow 0$

$$\cos \phi \rightarrow \cos \theta(\underline{\hat{Y}}, \underline{\hat{Z}})$$

To see this consider $\underline{dy} \cdot \underline{dz} = |\underline{dy}| |\underline{dz}| \cos \phi$

$$\Rightarrow \cos \phi = \frac{\underline{dy} \cdot \underline{dz}}{|\underline{dy}| |\underline{dz}|}$$

where $\underline{dy} \cdot \underline{dz} = (\underline{F} \underline{dY}) \cdot (\underline{F} \underline{dZ})$

$$= \underline{dY} \cdot \underline{F}^T \underline{F} \underline{dZ} = \underline{dY} \cdot \underline{C} \underline{dZ}$$

$$= \alpha^2 \underline{\hat{Y}} \cdot \underline{C} \underline{\hat{Z}} \quad \text{where } \underline{dY} = \alpha \underline{\hat{Y}} \text{ and } \underline{dZ} = \alpha \underline{\hat{Z}}$$

$$|\underline{dy}| = \sqrt{\underline{dy} \cdot \underline{dy}} = \alpha \sqrt{\underline{\hat{Y}} \cdot \underline{C} \underline{\hat{Y}}}$$

$$|\underline{dz}| = \sqrt{\underline{dz} \cdot \underline{dz}} = \alpha \sqrt{\underline{\hat{Z}} \cdot \underline{C} \underline{\hat{Z}}}$$

substituting into $\cos \phi = \frac{\underline{dy} \cdot \underline{dz}}{|\underline{dy}| |\underline{dz}|}$

$$\cos \phi = \frac{\underline{d\hat{Y}} \cdot \underline{C} \underline{d\hat{Z}}}{\sqrt{\underline{d\hat{Y}} \cdot \underline{C} \underline{d\hat{Y}}} \sqrt{\underline{d\hat{Z}} \cdot \underline{C} \underline{d\hat{Z}}}} \xrightarrow{\alpha \rightarrow 0} \cos \theta(\underline{d\hat{Y}}, \underline{d\hat{Z}})$$

Compute the shear $\gamma(\underline{\hat{Y}}, \underline{\hat{Z}}) = \theta(\underline{\hat{Y}}, \underline{\hat{Z}}) - \theta(\underline{\hat{Y}}, \underline{\hat{Z}})$

$\Rightarrow$ interpret components of $\underline{C}$

Components of $\underline{\underline{C}}$

$$\underline{\underline{C}} = C_{IJ} \underline{e}_I \otimes \underline{e}_J \Rightarrow C_{II} = \underline{e}_I \cdot \underline{\underline{C}} \underline{e}_I$$

I) Diagonal components:

$$C_{II} = \underline{e}_I \cdot \underline{\underline{C}} \underline{e}_I \quad (\text{no sum})$$

$$1^{\text{st}} \text{ Cauchy-Green: } \underline{\lambda}(\underline{Y}) = \sqrt{\underline{Y} \cdot \underline{\underline{C}} \underline{Y}}$$

$$\Rightarrow C_{II} = \lambda^2(\underline{e}_I) \quad \checkmark$$

II) Off-diagonal components

$$C_{IJ} = \underline{e}_I \cdot \underline{\underline{C}} \underline{e}_J$$

substitute into 2nd Cauchy-Green

$$\cos \theta(\underline{e}_I, \underline{e}_J) = \frac{\underline{e}_I \cdot \underline{\underline{C}} \underline{e}_J}{\sqrt{\underline{e}_I \cdot \underline{\underline{C}} \underline{e}_I} \sqrt{\underline{e}_J \cdot \underline{\underline{C}} \underline{e}_J}} = \frac{C_{IJ}}{\lambda(\underline{e}_I) \lambda(\underline{e}_J)}$$

$$\Rightarrow C_{IJ} = \lambda(\underline{e}_I) \lambda(\underline{e}_J) \cos \theta$$

shear: $\gamma(\underline{e}_I, \underline{e}_J) = \underbrace{\Theta(\underline{e}_I, \underline{e}_J)}_{\frac{\pi}{2}} - \Theta(\underline{e}_I, \underline{e}_J)$

$$= \frac{\pi}{2} - \Theta(\underline{e}_I, \underline{e}_J)$$

$$\Rightarrow \Theta(\underline{e}_I, \underline{e}_J) = \frac{\pi}{2} - \gamma(\underline{e}_I, \underline{e}_J)$$

substitute into C_{IJ}

$$\begin{aligned} C_{IJ} &= \lambda(\underline{e}_I) \lambda(\underline{e}_J) \cos\left(\frac{\pi}{2} - \gamma(\underline{e}_I, \underline{e}_J)\right) \\ &= \lambda(\underline{e}_I) \lambda(\underline{e}_J) \sin(\gamma(\underline{e}_I, \underline{e}_J)) \end{aligned}$$

Interpretation of components of $\underline{C}$:

$$C_{II} = \lambda^2(\underline{e}_I)$$

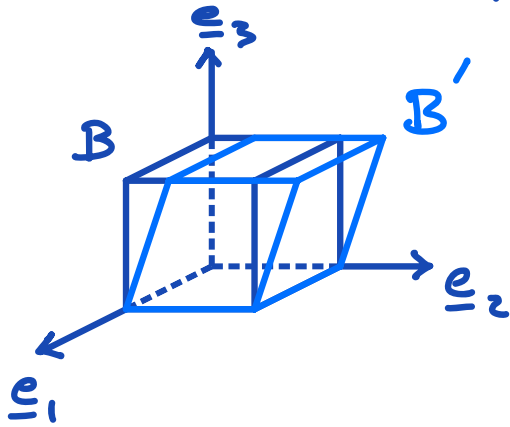
$$C_{IJ} = \lambda(\underline{e}_I) \lambda(\underline{e}_J) \sin \gamma(\underline{e}_I, \underline{e}_J) \quad (\text{no sum})$$

diagonal $\rightarrow$ square of stretches in coord. dir.

off diagonal $\rightarrow$ shear between coord. dir.

The components of $\underline{C}$ directly quantify stretch and shear unlike the components of $\underline{E}$.

Example: Simple shear



$$\underline{x} = \underline{\varphi}(\underline{X}) = \begin{bmatrix} \varphi_1 \\ \varphi_2 \\ \varphi_3 \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 + \alpha X_3 \\ X_3 \end{bmatrix} \quad \alpha > 0$$

"simple shear in $\underline{e}_2$ - $\underline{e}_3$ plane"

Deformation gradient:

$$\underline{F} = \nabla \underline{\varphi} = \begin{bmatrix} \varphi_{1,1} & \varphi_{1,2} & \varphi_{1,3} \\ \varphi_{2,1} & \varphi_{2,2} & \varphi_{2,3} \\ \varphi_{3,1} & \varphi_{3,2} & \varphi_{3,3} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow$ homogeneous deformation

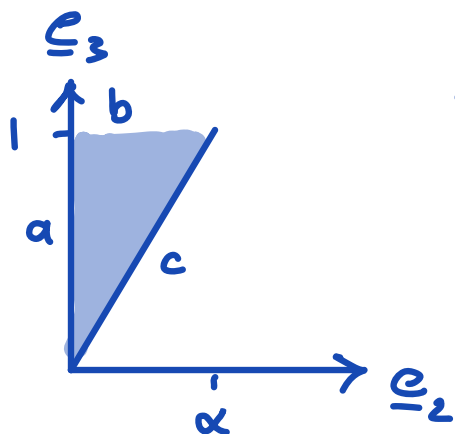
Cauchy-Green strain tensor:

$$\underline{C} = \underline{F}^T \underline{F} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & \alpha & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & 0 & 1 \end{bmatrix} = \underline{\underline{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & \alpha \\ 0 & \alpha & 1+\alpha^2 \end{bmatrix}}}$$

Stretches:

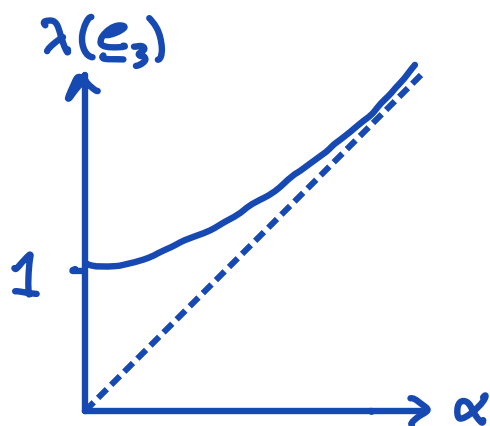
$$C_{11} = \lambda^2(\underline{e}_1) = 1 \quad \text{and} \quad C_{22} = \lambda^2(\underline{e}_2) = 1 \quad (\text{no stretch})$$

$$C_{33} = \lambda^2(\underline{e}_3) = 1 + \alpha^2 \Rightarrow \lambda(\underline{e}_3) = \sqrt{1 + \alpha^2}$$



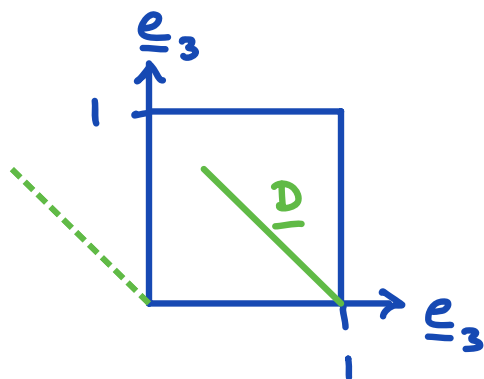
Pythagoras: $c = \sqrt{a^2 + b^2}$

$$\lambda(\underline{e}_3) = \frac{c}{a} = c = \sqrt{1 + \alpha^2} \quad a=1 \quad b=\alpha$$



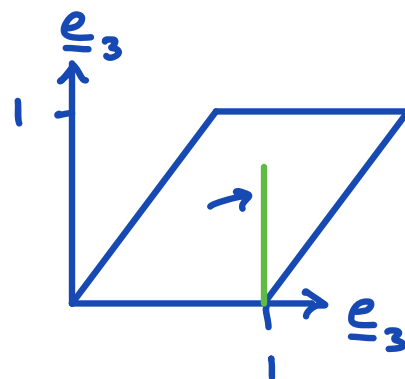
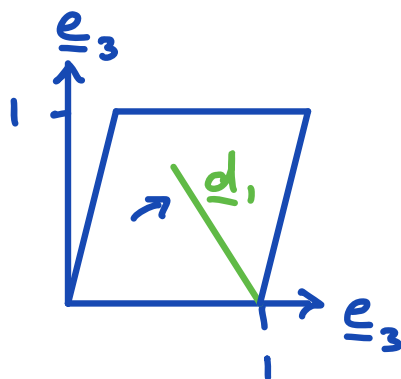
Stretch becomes linear
as vector is increasingly
horizontal.

What about non-coordinate directions?



$$\underline{D} = \frac{1}{\sqrt{2}}(\underline{e}_3 - \underline{e}_2)$$

$\lambda(\underline{D}) ? \Rightarrow 1^{\text{st}}$ Cauchy-Green SR



$$\lambda^2(\underline{D}) = \underline{D} \cdot \underline{C} \underline{D}$$

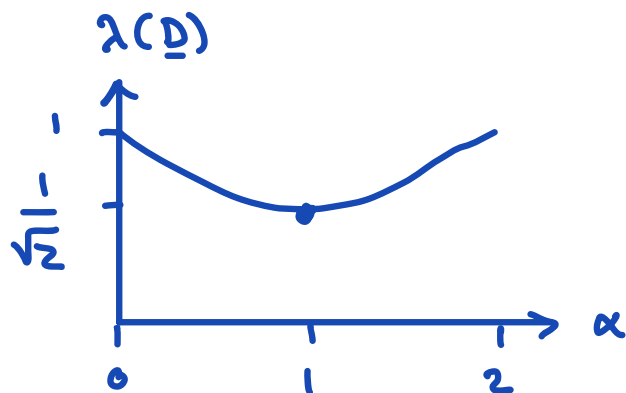
$$= \frac{1}{2} (\underline{e}_3 - \underline{e}_2) \cdot \underline{C} (\underline{e}_3 - \underline{e}_2)$$

$$= \frac{1}{2} (\underline{e}_3 \cdot \underline{C} \underline{e}_3 - \underline{e}_2 \cdot \underline{C} \underline{e}_3 - \underline{e}_3 \cdot \underline{C} \underline{e}_2 + \underline{e}_2 \cdot \underline{C} \underline{e}_2)$$

$$= \frac{1}{2} (C_{33} - 2C_{23} + C_{22}) \quad C_{23} = C_{32}$$

$$= \frac{1}{2} (1 + \alpha^2 - 2\alpha + 1)$$

$$\lambda^2(\underline{D}) = \frac{1}{2} \alpha^2 - \alpha + 1$$



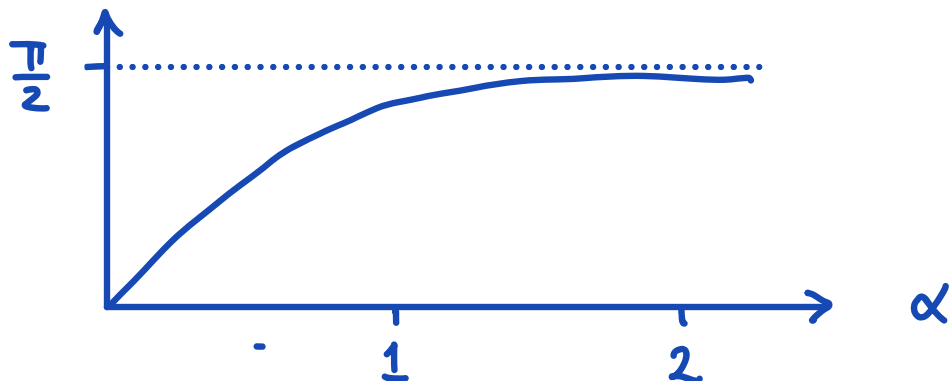
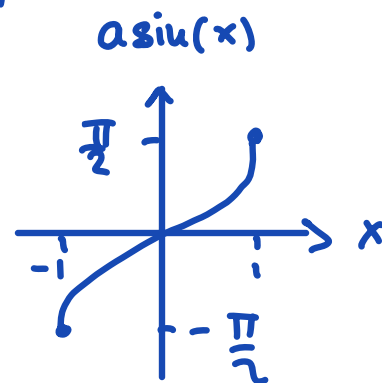
Shear: $C_{ij} \sim \sin(\gamma(\underline{e}_i, \underline{e}_j))$

$\Rightarrow C_{12} = C_{13} = 0$ no shear in these planes

$$C_{23} = \alpha = \cancel{\lambda(\underline{e}_2)} \cancel{\lambda(\underline{e}_3)} \sin(\gamma(\underline{e}_2, \underline{e}_3))$$

$$\gamma(\underline{e}_2, \underline{e}_3) = a \sin\left(\frac{\alpha}{\sqrt{1+\alpha^2}}\right)$$

$$\gamma(\underline{e}_2, \underline{e}_3)$$



$$\lim_{\alpha \rightarrow \infty} \frac{\alpha}{\sqrt{1+\alpha^2}} = 1$$

Principal stretches

$$(\underline{\underline{C}} - \mu_r \underline{\underline{I}}) \underline{\underline{v}}_r = \underline{\underline{0}}$$

$$\mu_r = \lambda_r^2$$

$$p(\mu) = \begin{vmatrix} 1-\mu & 0 & 0 \\ 0 & 1-\mu & \alpha \\ 0 & \alpha & 1+\alpha^2-\mu \end{vmatrix} = (1-\mu)^2(1+\alpha^2-\mu) - \alpha^2(1-\mu) = 0$$

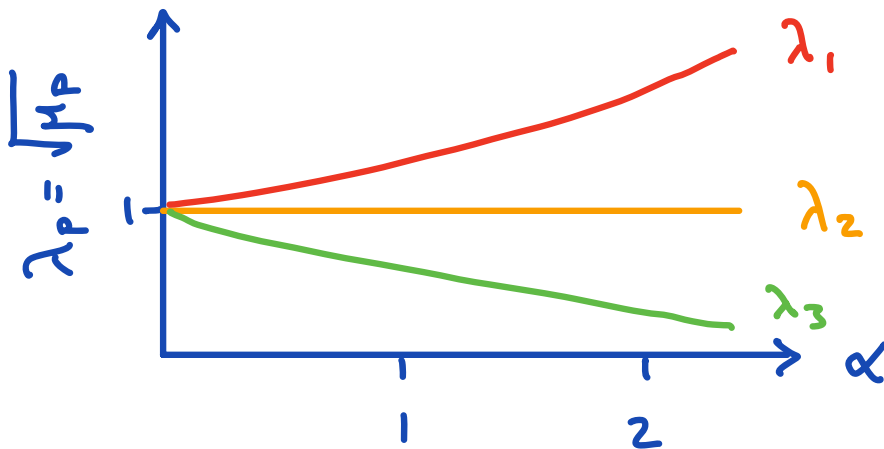
$$p(\mu) = (1-\mu) [(1-\mu)(1+\alpha^2-\mu) - \alpha^2] = 0$$

$$\Rightarrow \mu_2 = 1$$

$$(1-\mu)(1+\alpha^2-\mu) - \alpha^2 = 0$$

$$\mu^2 - (1+\alpha^2)\mu + 1 = 0$$

$$\Rightarrow \mu_{1/3} = 1 + \frac{1}{2} \alpha^2 \pm \frac{\alpha}{2} \sqrt{\alpha^2 + 4}$$



Principal directions:

$$(\underline{\underline{C}} - \mu_p \underline{\underline{I}}) \underline{v}_p = \underline{0}$$

$\Rightarrow$

$$\underline{v}_1 = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{2}(\alpha + \sqrt{1+\alpha^2}) \end{bmatrix}, \quad \underline{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \underline{v}_3 = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{2}(\alpha - \sqrt{1+\alpha^2}) \end{bmatrix}$$

note: $|\underline{v}_1| \neq 1$ $|\underline{v}_3| \neq 1 \Rightarrow$ normalize

Small deformation: $\alpha \ll 1$

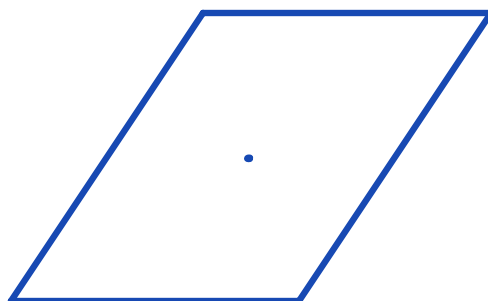
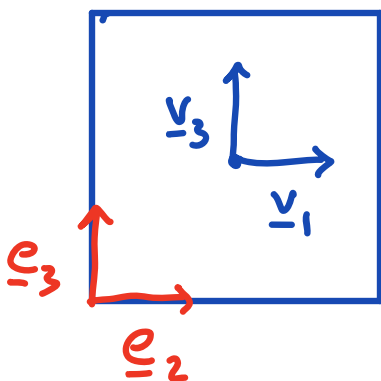
$$\lim_{\alpha \rightarrow 0} \underline{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\lim_{\alpha \rightarrow 0} \underline{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Large deformation: $\alpha \gg 1$

$$\lim_{\alpha \rightarrow \infty} \underline{v}_1 = \begin{bmatrix} 0 \\ 1 \\ \alpha \end{bmatrix}$$

$$\lim_{\alpha \rightarrow \infty} \underline{v}_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$



$$\underline{v}_1 = \frac{1}{\sqrt{1+\alpha^2}} \begin{bmatrix} 0 \\ 1 \\ \alpha \end{bmatrix}$$

$$\underline{v}_3 = \frac{1}{\sqrt{1+\alpha^2}} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

What are the extreme values of the stretch and their directions? $\Rightarrow$ eigenvalues & vectors

$$\begin{vmatrix} 1-\lambda^2 & \alpha & 0 \\ \alpha & 1+\alpha^2-\lambda^2 & 0 \\ 0 & 0 & 1-\lambda^2 \end{vmatrix} = 0$$

$$\begin{aligned} \lambda_1^2 &= 1 + \frac{\alpha^2}{2} + \alpha \sqrt{1 + \alpha^2/4} > 1 \\ \lambda_2^2 &= 1 \\ \lambda_3^2 &= 1 + \frac{\alpha^2}{2} - \alpha \sqrt{1 + \alpha^2/4} < 1 \end{aligned}$$

Principal directions:

$$[\underline{v}_1] = [\sqrt{1 + \alpha^2/4} - \alpha/2, 1, 0]$$

(not normalized)

$$[\underline{v}_2] = [0, 0, 1]$$

$$[\underline{v}_3] = [\sqrt{1 + \alpha^2/4} + \alpha/2, -1, 0]$$

$\Rightarrow \lambda_1$ is max stretch in dir $\underline{v}_1$
 λ_3 is min stretch in dir $\underline{v}_3$
 there is no stretch in dir $\underline{e}_3$

